Project Name	Project 2 - NIM - Automated Thematic Analysis
Online team meeting	https://fau.zoom-x.de/j/64229233763?pwd=MboXckMTe5aVIBRIwvO71OgqjXJ7zb.1
Production system (if any)	...
Test system (if any)	...
GitHub repository	https://github.com/amosproj/amos2026ss02-automated-thematic-analysis
GitHub feature board	https://github.com/orgs/amosproj/projects/98
GitHub imp-squared backlog	https://github.com/orgs/amosproj/projects/102
Position of T-shirt logo	Upper Torso Right Small
Link to logo for black T-shirt	https://github.com/amosproj/amos2026ss02-automated-thematic-analysis/blob/main/Deliverables/sprint-01/team-logo.svg
Link to logo for white T-shirt	https://github.com/amosproj/amos2026ss02-automated-thematic-analysis/blob/main/Deliverables/sprint-01/team-logo.svg
Additional materials	...
Team mailing list	oss-amos-proj2@lists.fau.de **(Emails sent here get quarantined by the TU Berlin Spam Filter)**

Last Name	First Name	GitHub User Name	Email Address
Kahvecioglu	Selin	slnkahveci	kahvecioglu@campus.tu-berlin.de
de Sa Filho	Fabiano	fdesafilho	fabiano.de.sa@fau.de
Ammann	Paul	Wecoreator	p.ammann@campus.tu-berlin.de
Schmidt	Alexander	alexanderschmi	a.schmidt.2@tu-berlin.de
Wendlandt	Julian	julianlucawendlandt-jpg	julian.luca.wendlandt@campus.tu-berlin.de
Müermann	Ole	ClimbTheGrind	ole.mueermann@fau.de
Klarer	Paul	paulKlarer	paul.linus.klarer@fau.de
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Annus	Mohammad	annus3	mohammad.annus@fau.de
Rasul	Hamid	hrasul	hamid.rasul@fau.de

		Product Owner					
#	Meeting Day	Review	Planning	Software Developer	Release Manager	Scrum Master	Comment
1	2026-04-15	Alexander Schmidt	Fabiano de Sa Filho	Everyone else	N/A	Christoph Mantsch	
2	2026-04-22	Alexander Schmidt	Fabiano de Sa Filho	Everyone else	Paul Klarer	Christoph Mantsch	
3	2026-04-29	Fabiano de Sa Filho	Alexander Schmidt	Everyone else	Paul Klarer	Christoph Mantsch	
4	2026-05-06	Alexander Schmidt	Fabiano de Sa Filho	Everyone else	Selin Kahvecioglu	Christoph Mantsch	
5	2026-05-13	Fabiano de Sa Filho	Alexander Schmidt	Everyone else	Ole Müermann	Christoph Mantsch	
6	2026-05-20	Alexander Schmidt	Fabiano de Sa Filho	Everyone else	Ole Müermann	Christoph Mantsch	
7	2026-05-27	Fabiano de Sa Filho	Alexander Schmidt	Everyone else	Hamid Rasul	Christoph Mantsch	Mid-term due
8	2026-06-03	Alexander Schmidt	Fabiano de Sa Filho	Everyone else	Selin Kahvecioglu	Christoph Mantsch	
9	2026-06-10	Fabiano de Sa Filho	Alexander Schmidt	Everyone else	Hamid Rasul	Christoph Mantsch	
10	2026-06-17	Alexander Schmidt	Fabiano de Sa Filho	Everyone else	Paul Ammann	Christoph Mantsch	
11	2026-06-24	Fabiano de Sa Filho	Alexander Schmidt	Everyone else	Mohammad Annus	Christoph Mantsch	
12	2026-07-01	Alexander Schmidt	Fabiano de Sa Filho	Everyone else	Mohammad Annus	Christoph Mantsch	
13	2026-07-08	Fabiano de Sa Filho	Alexander Schmidt	Everyone else	Paul Ammann	Christoph Mantsch	
14	2026-07-15	Alexander Schmidt	Fabiano de Sa Filho	Everyone else	Julian Wendlandt	Christoph Mantsch	Demo day!
15	2026-07-22	Fabiano de Sa Filho	Alexander Schmidt	Everyone else	Julian Wendlandt	Christoph Mantsch	Retrospective
Product owners, software developers, and Scrum Master are set and ideally don't change over time; the critical part is the Release Manager role you need to define here							

Goals	Having Fun
	Satisfy NIM
	Program an automated thematic analysis of interview transcripts
Meeting norms	Joining the meeting is mandatory
	Always be on time (up to 10 min late is fine if short message on discord)
	Follow the timeline
	Take notes in each meeting
	Happiness Index after Sprint retrospective
	Each meeting takes 1:30 hours
Working norms	Tell us if/when you are working on a task
	We encourage pair programming
Coordination norms	Self assign issues you are opening
	Do not force push
	Use PEP 8 as style (configure your IDEs)
	[Follow https://www.atlassian.com/git/tutorials/comparing-workflows/gitflow-workflow](https://www.atlassian.com/git/tutorials/comparing-workflows/gitflow-workflow)
	Once every workday check discord
	If someone is blocked share with the team
Communication norms	Criticism should be productive
	Language should be respectful at all time (written and oral)
Consideration norms	Discuss disagreement openly
	Follow the NDA and data privacy
	Try not to induce bias in our project
Cont. improvement norms	Finish work until tuesday evening 20:00 PM
	At the end of the meeting discuss happyness score / what could be done better / what was good
Rewards	Have a happy meeting
Sanctions	Write a poem saying you are sorry
	After multiple or serious offenses escalate to prof
Signatures	
Scrum Master	Christoph Mantsch
Product owner	Alexander Schmidt
Product owner	Fabiano de Sa Filho
Software developer	Ole Müermann
Software developer	Julian Wendlandt
Software developer	Hamid Rasul
Software developer	Paul Klarer

Software developer	Paul Ammann
Software developer	Selin Kahvecioglu
Software developer	Mohammad Annus

Product Vision	**Project Mission**
LLMs are a fantastic tool to interview research subjects at scale, which previously had to be done by human researchers. This method generates unstructured interview dialogues which must then be processed and analysed manually. The goal of this project is to use LLMs to also automate the analysis part, enabling policy researchers to feed their interview corpus to our tool and obtain not only the thematic analysis, but also relevant insights derived from the demographic data of the respondents. This way, we seek to bridge the gap between quantitative and qualitative policy research.	To engineer a secure, open-source tool that ingests thousands of subject interviews about a policy topic and produces an accurate thematic analysis breakdown. The core is a multi-stage LLM pipeline that extracts quotes, codes them into themes, clusters and ranks those themes, and cross-references them against respondent demographic dimensions. Results are surfaced through a lightweight frontend and exportable in structured formats. We also provide direct reference back to the corpus, so researchers can see which interviews most strongly manifest a particular theme or demographic combination.

Term	Definition
Corpus / Dataset	The LLM-led interviews on the economic impact of AI provided by the Industry Partner. Format: JSON
Demographic Data	Categorical variables describing each subject in the Corpus; has been partially sanitized by Industry partner, but must be handled with care. Absolutely should not be downloaded or uploaded to third parties. Format: CSV. Type: Numerical or Categorical Variables
Thematic Analysis	The qualitative method of identifying and analyzing patterns of meaning (themes) in a dataset. These themes should be followed by a short description and some quantitative measures of prevalence in the dataset.
Analysis Run	A single execution of the thematic analysis process based on a defined corpus, query, coding mode, and codebook. An analysis run produces outputs such as themes, tagged passages, and aggregated results.
Codebook	A structured collection of themes/codes used during thematic analysis. A codebook typically contains: - code/theme name - description - optional inclusion/exclusion criteria - optional examples
Interview	A textual dialogue between a research subject and an LLM about a research topic. It is provided by the industry partner in our case, and the interviews are about the Economic Impacts of AI in the job market.
Demographic Data Variables	Can be of two types: (1) Numerical: Continous variable like age, income. (2) Categorical: Classifies respondents into a category, such as gender, age bracket, level of schooling.
Coding Mode	The method by which the system determines and applies themes. The project currently includes three coding modes: - Fully automatic: The system consumes the corpus and produces the entire codebook without any human supervision, generating insights from the demographic data. - Semi-automatic: The system uses the corpus to produce a codebook, but allows the user to add, romove, and edit any of the themes. The system then procceds with the analyses using this edited codebook and receives natural language queries from the user, but does not modify the user-provided codebook in any way. - User-instructed: The system bypasses the codebook generation phase, and goes straight in to the natural language queries with the user provided codebook.
Corpus View	The product view that presents an aggregated overview of themes across the full corpus, including frequencies, quotes, and demographic breakdowns.
Interview View	The product view showing one full transcript, its highlighted themes, and the associated demographic information of the interviewee.
Tag	A thematic assignment attached to a specific passage in a transcript.
Query	A natural-language research question entered by the researcher to guide the thematic analysis.
Theme	A recurring pattern of meaning relevant to the research question, identified in one or more interviews.
Qualitative Analysis	A research method focused on interpreting non-numeric data, such as interview transcripts, to identify patterns, meanings, and themes.
Code	A short label, usually 5–12 words, that captures the central meaning of a quote or text chunk; it also has a description clarifying its scope.

Sprint #	Sprint goal
1	Meeting with industry partner, designing the logo, setting up the project
2	Ingesting the interview dataset provided by the industry partner, working on the Architecture Description document, and delivering a first prototype that successfully passes text through a basic LLM endpoint.
3	Creating a basic demo to be shown to Dr. Kaiser. Starting with simple views, database and llm communication
4	Creating the first version of the frontend, finishing up the basic demo of the backend. Adding a Readme and writting a proper Build script.
5	Closing up all features from previous sprints, finalizing the frontend so that we have a working demo for Industry Partner.
6	Working on the thematic analysis logic for the mid project release
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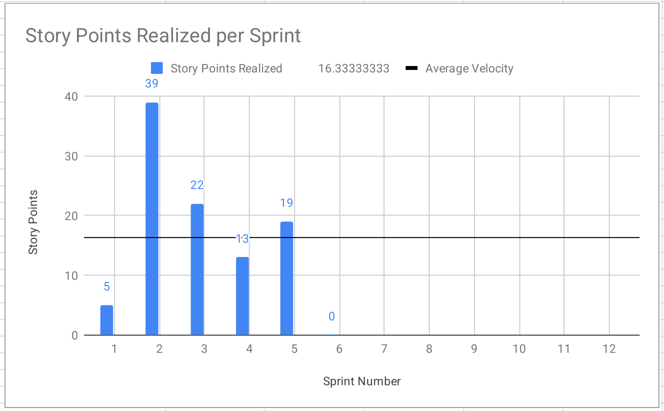
Sprint #	Story Points Realized	Story Points Planned	Average Planned	Average Realized
1	5	5	14.83333333	16.33333333
2	39	19	14.83333333	16.33333333
3	22	22	14.83333333	16.33333333
4	13	10	14.83333333	16.33333333
5	19	19	14.83333333	16.33333333
6	0	14	14.83333333	16.33333333
7			14.83333333	16.33333333
8			14.83333333	16.33333333
9			14.83333333	16.33333333
10			14.83333333	16.33333333
11			14.83333333	16.33333333
12			14.83333333	16.33333333
13			14.83333333	16.33333333
14			14.83333333	16.33333333
15			14.83333333	16.33333333
	PLEASE CREATE THE VELOCITY CHART ON A NEW TAB USING THE DATA FROM THIS TAB			



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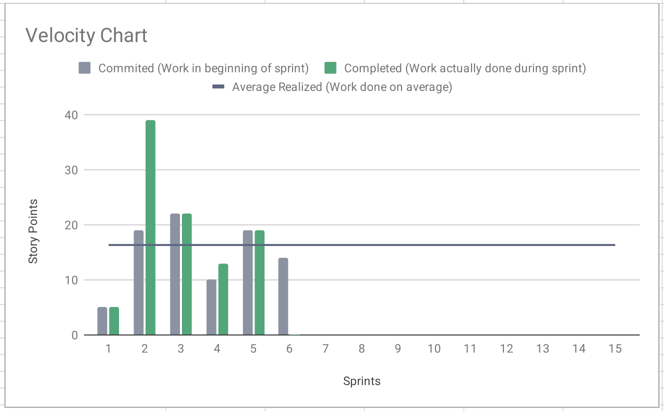
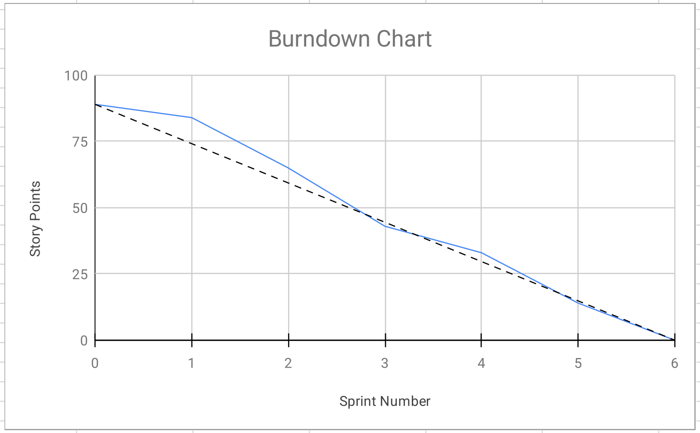
Scrum Master Charts



Sprint	Goal	Feature Name	Est. size	Est. remaining	Real size	Real remaining
Release						
Total			89		98	
Sprints						
0						
1	Meeting with industry partner, designing the logo, setting up the project		5	84	5	98
2	Ingesting the interview dataset provided by the industry partner, working on the Architecture Des		19	65	39	93
3	Creating a basic demo to be shown to Dr. Kaiser. Starting with simple views, database and llm c		22	43	22	54
4	Creating the first version of the frontend, finishing up the basic demo of the backend. Adding a F		10	33	13	32
5	Closing up all features from previous sprints, finalizing the frontend so that we have a working d		19	14	19	19
6	Working on the thematic analysis logic for the mid project release		14	0	0	0
Features						
1	**Meeting with industry partner, designing the logo, setting up the project**					
		Designing Project Logo	5		5	
2	**Ingesting the interview dataset provided by the industry partner, working on the Architecture Description document, and delivering a first prototype that successfully pass**					
		Select Open source model for Generation Phase	3		5	
		Create PDF of software architecture description	8		13	
		Software Bill of Material	8		13	
		Setting up the Repo with FastAPI, docker, etc	8		8	
3	**Creating a basic demo to be shown to Dr. Kaiser. Starting with simple views, database and llm communication**					
		Simple Data persistence Layer	5		5	
		View Theme Overview	8		8	
		Automated Database structure Setup	1		1	
		First LLM Request Setup	3		3	
		Plan Build Proccess	5		5	

Sprint	Goal	Feature Name	Est. size	Est. remaining	Real size	Real remaining
4	**Creating the first version of the frontend, finishing up the basic demo of the backend. Adding a Readme and writting a proper Build script.**					
		Upload Interview Transcript	3		3	
		Apply Codebook to interview	3		3	
		Add root README.md file	1		1	
		Implement Build Proccess	1		3	
		Setup Frontend for document Upload	2		3	
5	**Closing up all features from previous sprints, finalizing the frontend so that we have a working demo for Industry Partner.**					
		Setup Frontend	8		8	
		Link Transcripts and Demographic Data	5		3	
		Upload Predefined Codebook	3		3	
		Upload Demographic Data	3		5	
6	**Working on the thematic analysis logic for the mid project release**					
		Implement Automatic Codebook Generation	5			
		Demographic Data Upload Frontend	2			
		Demographic Data Viewer Page	2			
		Select Coding Mode Frontend	3			
		Export Codebook	2			



89	89
74	84
59	65
45	43
30	33
15	14
0	0

Speed 15 sp

Sprint	Goal	Feature Name	Est. size	Est. remaining	Real size	Real remaining
Release						
Total			0	0		
Sprints						
1			0	0	0	0
2			0	0	0	0
3			0	0	0	0
...				0		0
Features						
1						
2						
3						
		PLEASE CREATE THE BURNDOWN CHART ON A NEW TAB USING THE DATA FROM THIS TAB				

#	Feature Definition of Done	Sprint Release Definition of Done	Project Release Definition of Done
	New feature code has been documented, READMEs have been updated to reflect changes	Project builds, deploys, and tests successfully	Deployment process is tested and documented
	Unit tests for relevant logic are written and passing	All stories committed as “done” meet the Feature Definition of Done	User documentation has been updated
	Feature has been fully implemented	All completed features are integrated into the main branch	Architecture documentation and Bill of Materials is up to date
	Feature has been merged into the mainline	Documentation and READMEs reflect changes made in the sprint	The released product increment solves the intended user problem at a usable level
	All acceptance criteria were met	Incomplete items are clearly identified and moved/replanned	Performance is acceptable for the expected usage volume
	Product owner approved features	Retrospective completed	The user is able to install and use the application with minimal configuration
	Developers agreed to release	All implemented features are moved to the feature archive and their issues marked as closed	No critical or high bugs remain

Type	Link / reference

#	Context	Name	Version	License	Comment
1	Database	PostgreSQL	18	PostgreSQL License	Reliable open-source relational database for structured application data.
2	Database extension	pgvector	0.8	PostgreSQL License	Adds vector storage and similarity search directly inside PostgreSQL
3	Programming language / runtime	Python	3.11	Python Software Foundation License Version 2	Core runtime for backend
4	Database ORM / SQL toolkit	SQLAlchemy	2	MIT	Provides Python ORM and SQL toolkit support for database access and queries.
5	Backend API framework	FastAPI	0.136	MIT	Build easy python basd APIs
6	Data validation	Pydantic	2.13	MIT	Validates, parses, and serializes Python data models safely.
7	LLM application framework	LangChain	1.2	MIT	Build LLM workflows
8	NLP library	spaCy	3.8	MIT	Provides production-ready NLP features such as tokenization, NER, and pipelines. Contains e.g. BERT Implementations for NLP tasks like classification
9	ASGI server	Uvicorn	0.46	BSD-3-Clause	Runs the FastAPI backend
10	PostgreSQL async driver	asyncpg	0.31	Apache License 2.0	Provides fast asynchronous PostgreSQL connectivity for Python services.
11	LLM observability	Langfuse	4.5	MIT Expat for core; enterprise directories under separate ee/LICENSE	Tracks LLM traces, prompts, latency, costs, and evaluation data.
12	HTTP client	HTTPX	0.28	BSD-3-Clause	Provides sync and async HTTP calls to model APIs and backend services.
13	Testing framework	pytest	9	MIT	Automated Testing
14	Environment configuration	python-dotenv	1.2	BSD-3-Clause	Loads local environment variables from .env files during development.
15	Tokenization	tiktoken	0.12	MIT	Counts and encodes tokens. Used by LangChain for Textsplitting.
16	LLM frontend prototyping	Streamlit	1.56	Apache License 2.0	Quickly builds Python-based web apps, dashboards, and chat prototypes for LLM applications. Used for protoyping & early previews.
17	Large language model	Gemma	Gemma 4 31B	Apache License 2.0	Candidate self-hosted open model. More details in the model selection document. Final model decision is still outstanding.

#	Context	Name	Version	License	Comment
18	Large language model	Qwen	Qwen3.5-35B-A3B	Apache License 2.0	Candidate self-hosted open model. More details in the model selection document. Final model decision is still outstanding.
19	Large language model	Mixtral	Mixtral 8x7B	Apache License 2.0	Candidate self-hosted open model. More details in the model selection document. Final model decision is still outstanding.
20	LLM gateway / provider abstraction	LiteLLM	1.83	MIT for OSS package; enterprise package/features under separate commercial license	Provides a unified OpenAI-compatible interface for calling multiple LLM providers.
21	Containerization / deployment	Docker Engine	29	Apache License 2.0	Packages and runs the application stack consistently across environments.
22	Web framework	Flask	3.1	BSD-3-Clause	Server-side web framework for the production frontend
23	Template engine	Jinja2	3.1	BSD-3-Clause	Template engine used with Flask to generate server-rendered HTML views
24	Chat / users	OpenWebUI	0.9	Open WebUI License	chat-style interaction and user management. Current OpenWebUI versions use the Open WebUI License, which is based on BSD-3-Clause but adds a branding / attribution restriction

#	Context	Name	Version	License	Comment

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	First Name	Value					
Kahvecioglu	**Selin**	2		**2.00**	**OK**		
de Sa Filho	**Fabiano**	N/A					
Ammann	**Paul**	2					
Wendlandt	**Julian**	2		0	No size		
Müermann	**Ole**	2		1	Trivial size		
Klarer	**Paul**	2		2	Small size		
				3	Medium size		
Annus	**Mohammad**	2		5	Large size		
Rasul	**Hamid**	2		8	Very large size		
				13	Too large (size)		
How to play planning poker							
1. Everyone type their number into their value field, don't hit return yet							
2. Someone, perhaps a product owner, count down 3.. 2.. 1..							
3. Then, everyone hit return to submit their value							